

Monday	-	3	10	17	24
Tuesday	-	4	11	18	25
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11 week

Q1.

- (a) (i) $W_0 = 0$, the asset's cumulative stochastic return component starts at zero today.
- (ii) continuous paths: Asset prices evolve continuously without sudden, discrete macro-jumps.
- (iii) Independent increments: Past market price movements provide zero info. about future price movements.
- (iv) stationary Gaussian increments ($W_t - W_s \sim N(0, t-s)$)
Price uncertainty grows linearly over time, and returns over an interval are normally distributed.
- (b) Assume $s \leq t$, we expand W_t into independent components.

$$W_t = W_s + (W_t - W_s)$$

$$\text{Cov}(W_s, W_t) = \text{Cov}(W_s, W_s + (W_t - W_s)) = \text{Cov}(W_s, W_s) + \text{Cov}(W_s, W_t - W_s)$$

Since W_s & $W_t - W_s$ are independent, their covariance is 0.

$$\text{Cov}(W_s, W_t) = \text{Var}(W_s) + 0 = s$$

By symmetry, for $t \leq s$, we get,

$$\text{Cov}(W_s, W_t) = \min(s, t)$$

07

MARCH
Tuesday

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c) Given $Q^n = \sum_{k=0}^{n-1} (W_{t_{k+1}} - W_{t_k})^2$, let $\Delta t_k = t_{k+1} - t_k$

Since $(W_{t_{k+1}} - W_{t_k}) \sim N(0, \Delta t_k)$, its expected squared value is its variance.

$$E[(W_{t_{k+1}} - W_{t_k})^2] = \Delta t_k$$

By linearity of expectation;

$$E[Q^n] = \sum_{k=0}^{n-1} E[(W_{t_{k+1}} - W_{t_k})^2] = \sum_{k=0}^{n-1} \Delta t_k$$

d) $\text{Var}(Z^2) = 2\sigma^4$ for $Z \sim N(0, \sigma^2)$. We have

$\text{Var}((W_{t_{k+1}} - W_{t_k})^2) = 2(\Delta t_k)^2$. Due to independent increments,

$$\begin{aligned} \text{Var}(Q^n) &= \sum_{k=0}^{n-1} \text{Var}((W_{t_{k+1}} - W_{t_k})^2) \\ &= 2 \sum_{k=0}^{n-1} (\Delta t_k)^2 \leq 2 \|T\| \sum_{k=0}^{n-1} \Delta t_k \\ &\leq 2 \|T\| T \end{aligned}$$

As $\|T\| \rightarrow 0$, $\text{Var}(Q^n) \rightarrow 0$. Because $E[Q^n] = T$ and $\text{Var}(Q^n) \rightarrow 0$, $Q^n \rightarrow T$ in L^2 .

It's remarkable that every individual realization of a Brownian path is completely random and jagged, yet the sum of its squared steps yields an identical, perfectly deterministic no. (T).

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Q2.(a) For $f(t, x_t)$;

$$df(t, x_t) = \left(\frac{\partial f}{\partial t} + \mu(t, x_t) \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2(t, x_t) \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma(t, x_t) \frac{\partial f}{\partial x} dW_t$$

$\Rightarrow \frac{\partial f}{\partial t} dt$: Explicit time-decay drift of f^n .

$\mu \frac{\partial f}{\partial x} dt$: First-order deterministic state drift.

$\frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} dt$: Ito's 2nd order correction term.

$\sigma \frac{\partial f}{\partial x} dW_t$: Stochastic diffusion component.

$\Rightarrow$ Ordinary chain rule is modified as, stochastic increments satisfy $(dW_t)^2 = dt$, meaning 2nd order spatial derivatives generate 1st order temporal (dt) impacts.

(b) Let $f(S_t) = \ln(S_t)$. $\frac{\partial f}{\partial t} = 0$, $\frac{\partial f}{\partial S} = \frac{1}{S_t}$, $\frac{\partial^2 f}{\partial S^2} = -\frac{1}{S_t^2}$

$\Rightarrow$ Applying Ito's lemma,

$$d(\ln S_t) = \left(0 + (\mu S_t) \frac{1}{S_t} + \frac{1}{2} (\sigma S_t)^2 \left(-\frac{1}{S_t^2} \right) \right) dt + (\sigma S_t) \frac{1}{S_t} dW_t$$

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$$d(\ln S_t) = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t$$

Integrating from 0 to T,

$$\frac{S_T}{S_0} = \left(\mu - \frac{\sigma^2}{2} \right) T + \sigma W_T \Rightarrow S_T = S_0 e^{(\mu - \frac{\sigma^2}{2})T + \sigma W_T}$$

② ⇒ The $-\frac{\sigma^2}{2}$ term represents a downward adjustment to the geometric growth rate caused by volatility drag (compounding over fluctuating returns reduces the median path).

$$\begin{aligned} E(S_T) &= S_0 e^{(\mu - \frac{\sigma^2}{2})T} E(e^{\sigma W_T}) = S_0 e^{(\mu - \frac{\sigma^2}{2})T} e^{\frac{\sigma^2 T}{2}} \\ &= S_0 e^{\mu T} \end{aligned}$$

$$\begin{aligned} \text{If we omit correction, } E(S_{10}) &= 100 e^{0.07 \times 10} = 100 e^{0.7} \\ &= 100 e^{0.7} \approx 194.77 \\ &= 100 e^{0.9} \approx 245.96 \end{aligned}$$

$$\text{True Expectation} = 100 e^{0.7} \approx 201.38$$

$$\text{Absolute error} \approx 44.58 \quad (22.1\% \text{ overestimation error})$$

③ Speed ⇒ Higher σ increases the variance and skewness causing the right-hand tail of S_t to lengthen dramatically while shifting the peak (mode) leftward.

④ Option payoff ⇒ $E[(S' - K)^+]$: Higher volatility increases the expected payoff because call options have uncapped upside potential from the wider right tail, while down losses are strictly limited to 0.

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MARCH
Friday

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11. week

83.

(a) Drift $\rightarrow (-\theta x_t)$, Diffusion (σ)

(2) Mechanism - If $x_t > 0$, drift is negative, dragging the process down. If $x_t < 0$, drift is +ve, pulling it up.

(3) When x_t is highly +ve or -ve, the linear drift magnitude $-\theta x_t$ dominates overwhelmingly over the random diffusion shocks, enforcing an aggressive, near-deterministic return to 0.

(b)
$$x_{t+k+1} = x_{t+k} - \theta x_{t+k} \Delta t + \sigma \Delta W_k$$

Stability condition - $\Delta t < 2/\theta$. If Δt exceeds the limit, the deterministic update overcorrects, resulting in expanding, explosive oscillations.

Stationary Distribution

↳ Regardless of starting value x_0 , as $T \rightarrow \infty$, the prob. distribution of x_T settles into a steady state normal profile $N(0, \sigma^2/2\theta)$

↳ It accurately mirrors economic realities: interest rates do not grow infinitely nor collapse permanently to 0, they oscillate around a structural baseline.

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MARCH
Saturday

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f

speed: higher σ forces a steeper, much faster decay back to 0 from initial value X_0

Width: higher σ narrows the long-run variance (keeping the paths bounded closer to 0).

g

Strong Convergence (order $\frac{1}{2}$) - measures the avg. error of individual simulated path trajectories matching the actual exact Brownian path.

Weak Convergence (order $\frac{1}{2}$) - measures how accurately the overall prob. distribution matches the exact distribution.

Option pricing - Since option pricing relies solely on calculating expected values across distribution, matching aggregate distribution profile via weak convergence is sufficient.